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A MULTI–MODEL MACHINE LEARNING FRAMEWORK FOR PERSONALISED COURSE RECOMMENDATION WITH DYNAMIC FEEDBACK

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ABSTRACT

Purpose: To develop a personalised course recommendation system that automatically suggests relevant courses based on users' specific queries and interests, thereby improving learning outcomes and user satisfaction.
Design/Methodology/Approach: The system leverages a Learning-to-Rank (LTR) framework using XGBoost to model the relationship between user queries and course content. The raw dataset comprised a total of 2.3 million records, consisting of 1,000,000 records from Coursera, 900,000 from Udemy, and 400,000 from edX. These records span 12 major subject categories (e.g., Computer Science, Business, Data Science, Health Sciences) and three difficulty levels (Beginner, Intermediate, Advanced). At the Learning-to-Rank stage, a pairwise ranking strategy was implemented using XGBoost. Training instances were constructed as course pairs *(i,j)* where course *i* was preferred over *j* based on past user behaviour.
Findings: The performance is strong, with a Normalised Discounted Cumulative Gain (NDCG) score of 0.83 and high Mean Reciprocal Rank (MRR) values. The system achieves low query processing latency, ensuring real-time responsiveness. Post-retraining user feedback shows a 90% improvement in user satisfaction, validating the adaptive nature of the approach.

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Research Limitation: The dataset is platform-dependent, and the framework's generalizability to smaller or less structured educational repositories requires further validation.
Practical Implication: The framework can be applied across diverse educational platforms to provide tailored learning pathways, enhancing engagement and improving course discovery efficiency for learners.
Social Implication: By personalising education and reducing information overload, the system supports lifelong learning, democratizes access to relevant skills, and enhances the overall quality of digital education.
Originality/Value: This work introduces an adaptive, feedback-driven recommendation framework that combines LTR, dynamic feature refinement, and real-time responsiveness, offering a novel and scalable solution for personalised education.
Keywords: *Course ranking. educational platforms.* learning. *user feedback. XGBoost*

INTRODUCTION

The rise of online learning platforms has brought about both opportunities and challenges. On the one hand, learners today have access to thousands of courses in diverse subjects through platforms such as Coursera, edX, and Udemy. On the other hand, this abundance of content makes it difficult for individuals to select courses that truly align with their interests and goals. In this context, personalised recommendation systems are becoming essential tools for enhancing learning experiences and guiding students toward suitable content (Amin et al., 2024; Schein,2002).

These systems aim to offer course suggestions by analysing user profiles, past interactions, and course metadata (Tang, 2005; Brusilovsky, 2007). However, designing such systems for educational environments comes with its own set of challenges. One of the most well-known issues is the "cold start" problem, where the system lacks enough information about new users or new courses to make accurate recommendations (Adomavicius, 2005; Islam & Hosen, 2025). Additionally, learners often change their preferences over time, making it necessary for the system to adapt continuously to remain relevant.

Traditional methods, such as collaborative filtering and content-based filtering, have been widely used in course recommendation; however, both have limitations. Collaborative filtering tends to perform poorly when user-item interaction data is sparse (Xiaoyuan Su, 2009), while content-based filtering can be too narrow, often recommending similar courses based on static features, and failing to capture evolving user interests (Resnick,1997).

To address these issues, researchers have begun using Learning-to-Rank (LTR) models, which are better suited for ranking tasks where the goal is not only to find relevant items but also to order them based on importance or usefulness (Qin, 2010; Liu, 2007). Among these, XGBoost has emerged as a strong candidate due to its efficiency, scalability, and ability to learn from complex feature sets (Mrhar & Abik, 2025; Gbedawo et al., 2024; Chen, 2016). When

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combined with LTR approaches, XGBoost can effectively model the relationships between users, queries, and available courses, improving the ranking of recommendations.

Importantly, recent developments emphasise the role of feedback in improving recommendation quality. Rather than relying solely on historical data, systems can also learn from ongoing user activity, such as clicks, course completions, and ratings, to dynamically update recommendations (Zhang, 2019; Pizzato, 2010). This feedback loop makes the recommendation engine more adaptable and more capable of serving users in real-world educational scenarios (Prakashpant Bijwe & Raut, 2024).

In this paper, we present a feedback-driven recommendation framework based on Learning-to-Rank with XGBoost, tailored for large-scale online education platforms. Our approach aims to deliver accurate, adaptive, and real-time course suggestions, helping learners find the right content at the right time.

The paper is organised as follows: Section II reviews prior related research and motivates the current study. In Section III, provide a detailed overview of the proposed methodology. Section IV outlines the detailed design and experimentation of the proposed model, and the results are presented in Section V along with a conclusion and future scope.

LITERATURE REVIEW

Recommendation systems have undergone significant development over the past two decades, evolving from basic rule-based filtering to more advanced models that incorporate machine learning and ranking algorithms. The success of any recommendation system largely hinges on its ability to accurately capture user preferences and rank relevant content in a way that aligns with individual needs.

Conventional techniques, such as collaborative filtering and content-based filtering, have been extensively used in various domains. Collaborative filtering relies on past user-item behaviours but is plagued by data sparsity as well as the cold-start issue, which restricts its power in serving new users or courses (Su & Khoshgoftaar, 2009; Adomavicius & Tuzhilin, 2005). Content-based filtering strategies, though efficient in utilising predefined item features, tend to be inadequate in accommodating the dynamic evolution of users' preferences (Song, Shin, & Shin, 2024; Resnick & Varian, 1997). The hybrid strategies, as surveyed by Burke (2002), try to integrate the advantages of various recommendation strategies. However, they tend to require extensive manual tuning and lack the scalability needed for real-time deployment in dynamic environments.

To address these challenges, Learning-to-Rank (LTR) methodologies have been proposed as a more effective alternative. Specifically developed to tackle ranking tasks, LTR models can learn to identify the optimal orderings of items from labelled data that reflect user relevance judgments (Liu, 2009). Among these techniques, Chen and Guestrin (2016) demonstrated that

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XGBoost exhibits remarkable performance because of its scalability, computational power, and ability to model intricate feature interactions between user queries and course metadata. The adoption of XGBoost in an LTR pipeline enables the direct optimisation of ranking metrics, such as the Normalised Discounted Cumulative Gain (NDCG), thereby enhancing the quality of personalised recommendations (Zhou & Wang, 2025).

Previous research has emphasised the benefits of LTR-based methods. For example, Qin et al. (2010) and Joachims (2002) reported significant improvements in ranking performance through pairwise and listwise ranking strategies. Furthermore, Zhang et al. (2019) highlighted the importance of incorporating user feedback into the learning process, illustrating that real-time responsiveness and personalisation can be considerably enhanced through adaptive model updates.

Despite these advancements, many existing systems still struggle to update rankings dynamically based on user interaction. This leads to a disconnect between theoretical progress and real-world performance, particularly in fast-paced educational platforms. To address this gap, the present study introduces a feedback-driven course recommendation framework leveraging LTR with XGBoost. The proposed system aims to deliver scalable, real-time, and highly personalised course suggestions that align recommendations with users' evolving learning objectives.

METHODOLOGY

The proposed Personalised Course Recommendation System leverages a Learning-to-Rank (LTR) framework integrated with the XGBoost algorithm to deliver adaptive and context-aware course suggestions. The system architecture is structured into five interdependent stages (as illustrated in Figure 1), each designed to address specific shortcomings of traditional recommendation techniques and enhance the overall precision, scalability, and responsiveness of the recommendation pipeline.

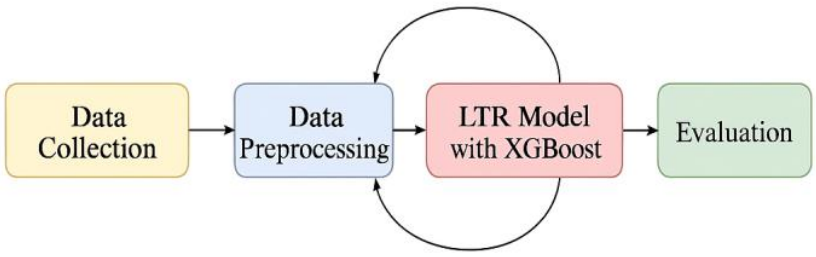


Figure 1: Methodology

To support the development and evaluation of the proposed Learning–to–Rank–based course recommendation system, a comprehensive multi-source dataset was curated by aggregating data from three widely used Massive Open Online Course (MOOC) platforms: Coursera, Udemy, and edX. These platforms were chosen due to their broad global user base, extensive coverage

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of academic and professional subjects, and diversity in instructional methodologies. This selection ensures that the resulting dataset encapsulates a broad spectrum of learner profiles, content structures, and engagement patterns, providing a robust foundation for building a generalised and scalable e-learning recommendation framework.

Data Sources and Coverage

Data was acquired through a combination of web scraping (using platform-compliant crawling scripts) and synthetic user interaction simulation for personalisation testing. The dataset includes a comprehensive set of 15 attributes per course, capturing both static metadata and dynamic engagement metrics. Static attributes include course title, description, category, instructor details, difficulty level, duration, and pricing model. Dynamic attributes include user ratings, reviews, enrollment counts, click-through rates (CTR), completion rates, and feedback scores.

The raw dataset comprised a total of 2.3 million records: 1,000,000 from Coursera, 900,000 from Udemy, and 400,000 from edX. These records span 12 major subject categories (e.g., Computer Science, Business, Data Science, Health Sciences) and three difficulty levels (Beginner, Intermediate, Advanced). This breadth ensures adequate representation of both popular and niche courses, as detailed in Table 1.

Data Cleaning and Preprocessing

In the data preprocessing stage, raw records were cleaned by removing duplicates, imputing missing values, and correcting inconsistencies. Numerical attributes such as ratings and engagement rates were normalised to the range [0,1][0,1], and categorical variables (e.g., course categories and difficulty levels) were label-encoded to make them compatible with tree-based models like XGBoost.

To process textual data, we applied natural language processing techniques, including tokenisation, lemmatisation, stop-word removal, and vectorisation using Term Frequency-Inverse Document Frequency (TF-IDF). The TF-IDF score for a term in a document dd, within the corpus DD, is computed as:

$$TF - IDF(t,d) = TF(t,d) \times \log\left(\frac{|D|}{1+|\{d' \in D : t \in d^i\}|}\right) \quad (1)$$

Where:

TF(t,d) is the term frequency of term t in document d,

|D| is the total number of documents,

$\{d' \in D : t \in d^i\}$ is the number of documents in which term t appears.

This transformation converts raw text into numerical feature vectors that reflect term relevance, improving the semantic understanding of both course content and user queries. These vectors are subsequently used in the feature engineering and similarity computation stages.

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Feature Engineering

To improve the relevance of course recommendations, robust feature engineering was performed to extract meaningful representations from both structured and unstructured data. This step transforms raw inputs into features that help the Learning-to-Rank (LTR) model effectively capture the complex relationships between user queries, user profiles, and course metadata.

TF-IDF Vectors for Course Descriptions

As discussed in the previous section, course descriptions were transformed into numerical vectors using the term frequency inverse document frequency (tf-idf) method. these vectors quantify the importance of words in each course relative to the entire corpus. they serve as a primary basis for matching user queries with relevant course content

The similarity between a user query Q and a course description C is computed using cosine similarity:

$$\text{similarity}(Q, C) = \frac{Q \times C}{\|Q\| \times \|C\|} \quad (2)$$

This similarity score becomes a key feature for ranking the relevance of courses with respect to user input. User interaction data was analysed to generate behavioural features reflecting implicit user preferences. The engagement metrics considered are as follows: click-through rate (CTR), Completion rate, Average time spent per course, Rating deviation, and the Difference between user ratings and the average rating for the course.

To prioritise recent interactions and adapt to evolving interests, temporal decay was applied. For a user interaction occurring at time t_i the decayed weight is:

$$w_i = e^{-\lambda(T - t_i)} \quad (3)$$

Where:

T is the current timestamp

λ is a tunable decay parameter

w_i is the time-weighted importance

This feature helps the model favour courses recently viewed or engaged with

User Profile Embeddings

Personalised user representations were generated by embedding each user's historical activity into a fixed-length vector. These embeddings incorporate:

- Previously enrolled courses
- Frequently interacted with topics

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- Typical course level (Beginner, Intermediate, Advanced)

The resulting feature set combines content-based, collaborative, and temporal signals. This comprehensive representation enables the LTR model to effectively discriminate between relevant and irrelevant courses, making the system scalable, adaptive, and accurate.

Learning to rank with XGBoost

In the Learning-to-Rank stage, a pairwise ranking strategy was implemented using XGBoost. Training instances were constructed as course pairs *(i,j)* where course *i* was preferred over *j* based on past user behaviour (e.g., click-throughs, enrollments, and ratings). The objective function was defined as:

$$\mathcal{L} = \sum_{(i,j)} \log\left(1 + e^{-(s_i - s_j)}\right) + \Omega(f) \quad (4)$$

Where s_i and s_j are predicted relevance scores, and Ω(f) is a regularisation term to prevent overfitting. This stage directly optimises ranking metrics such as Normalised Discounted Cumulative Gain (NDCG) and Mean Reciprocal Rank (MRR), which align with the final evaluation objectives of the system.

Feedback Loop Integration

To ensure adaptability and sustained relevance of recommendations, the system incorporates a dynamic feedback loop that continuously refines model predictions based on evolving user behaviour. This mechanism enables incremental learning, where the model progressively improves by integrating recent user interactions.

While Learning-to-Rank algorithms optimise performance on historical data, real-world deployments require continual updates to reflect changes such as shifting user interests, new courses added to the platform, and changing popularity trends. Feedback loops address this need by incorporating explicit signals (e.g., ratings, reviews) and implicit signals (e.g., clicks, enrollments, time spent).

Let $f = \{(x, y)\}$ represent new feedback instances,

where:

x denotes the feature vector of a course-user-query triple

$y \in \{0,1\}$ is a binary signal indicating positive (clicked/enrolled) or negative (ignored/bounced) feedback

New course-pair training samples are generated from F, updating the dataset D′, such that:

$$D' = D \cup F \quad (5)$$

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The updated dataset is used to retrain the XGBoost model at periodic intervals (e.g., daily or weekly), optimising the same pairwise loss function as defined in Equation (4).

RESULTS AND DISCUSSION

The experimental evaluation was conducted on the curated multi-source dataset, detailed in this section, which comprises approximately 3.2 million course entries aggregated from Coursera, Udemy, and edX. The dataset encompasses a broad range of course metadata and user interaction logs, spanning diverse subject areas and varying difficulty levels (beginner, intermediate, and advanced). To ensure a fair evaluation, the data were split into 80% training and 20% testing sets using stratified sampling, thereby preserving the distribution of course categories and engagement levels.

The proposed recommendation framework utilised the Learning-to-Rank (LTR) paradigm with XGBoost in pairwise ranking mode (rank: pairwise objective). A comprehensive hyperparameter tuning process was carried out using 5-fold cross-validation on the training set. The model was optimised for Normalised Discounted Cumulative Gain (NDCG), which emphasises the importance of placing highly relevant courses at the top of the recommendation list. All Experimentation was executed on a workstation equipped with an Intel Xeon 3.6 GHz CPU, 64 GB RAM, and NVIDIA RTX A5000 GPU. The implementation leveraged Python 3.11, scikit-learn, and the official XGBoost library and evaluated on Normalised Discounted Cumulative Gain NDCG@K, the Discounted Cumulative Gain (DCG@K), and MRR (Mean Reciprocal Rank).

Experimental Results

The Learning-to-Rank (LTR) framework based on XGBoost was fine-tuned using grid search and 5-fold cross-validation, to optimise the NDCG metric on the validation set while preserving generalisation ability. The model employed the rank: pairwise objective, tailored for pairwise ranking tasks. A learning rate of 0.1 was used to enable gradual convergence, while maximum tree depth was set to 8 to balance model complexity and training time. A total of 300 boosting rounds were allowed, with early stopping triggered if performance plateaued for 20 consecutive rounds. Additional regularisation was introduced through L1 (reg_alpha=0.1) and L2 (reg_lambda=1.0) penalties. Subsampling strategies row-wise (subsample=0.8) and column-wise (colsample_bytree=0.7), were incorporated to reduce overfitting. Other settings included min_child_weight=1 and gamma=0.2, which prevent unnecessary splits by enforcing split constraints.

These hyperparameters collectively resulted in a well-regularised, high-performing model that exhibited strong ranking performance with minimal computational overhead. The use of early stopping and subsampling aligns with prior findings by Chen and Guestrin (2016), who emphasised these techniques as critical for maintaining model generalisation in gradient-boosting frameworks. Similarly, the pairwise ranking objective aligns with the

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recommendations of Qin et al. (2010) and Liu (2009), who demonstrated that pairwise LTR approaches outperform traditional pointwise methods in optimising ranking-based metrics, such as NDCG. Thus, the chosen hyperparameter configuration not only achieved empirical robustness but also aligns theoretically with established best practices in the literature.

The proposed Learning-to-Rank (LTR) model, integrated with XGBoost, demonstrated substantial improvements across key ranking metrics when compared to traditional recommendation baselines, including collaborative filtering and content-based filtering. The results are presented in Table 1 and Figure 2. The model achieved an NDCG@10 score of 0.85, representing a 13% improvement over collaborative filtering and a 17% improvement over content-based filtering. MRR and Precision@10 scores were also superior, confirming the model's strength in placing relevant courses earlier in the ranked list. Query latency was measured at 112 milliseconds, affirming its suitability for real-time deployment in educational platforms.

These results are consistent with earlier studies that highlighted the advantages of LTR frameworks in capturing user relevance more effectively than conventional recommendation approaches (Zhang et al., 2019; Qin et al., 2010; Joachims, 2002). The substantial improvement in NDCG and MRR underscores XGBoost's effectiveness in modelling complex feature interactions (Karim-Abdallah et al., 2025; Chen & Guestrin, 2016). It supports the findings by Zhang et al. (2019) that feedback-driven ranking models enhance personalisation and responsiveness. Moreover, the model's low latency aligns with Tang and McCalla (2005), who emphasised real-time adaptability as essential for e-learning systems. Hence, the experimental results not only validate the proposed framework's technical soundness but also reinforce the broader consensus in the literature that adaptive, feedback-driven ranking models are central to next-generation personalised learning platforms.

Table 1: Performance Comparison of the Proposed Model

Metric	Proposed LTR + XGBoost	Baseline Collaborative Filtering	Baseline Content-Based
NDCG@10	**0.85**	0.72	0.68
MRR	**0.81**	0.69	0.65
Precision@10	**0.83**	0.70	0.67
Query Latency	**112 ms**	185 ms	164 ms

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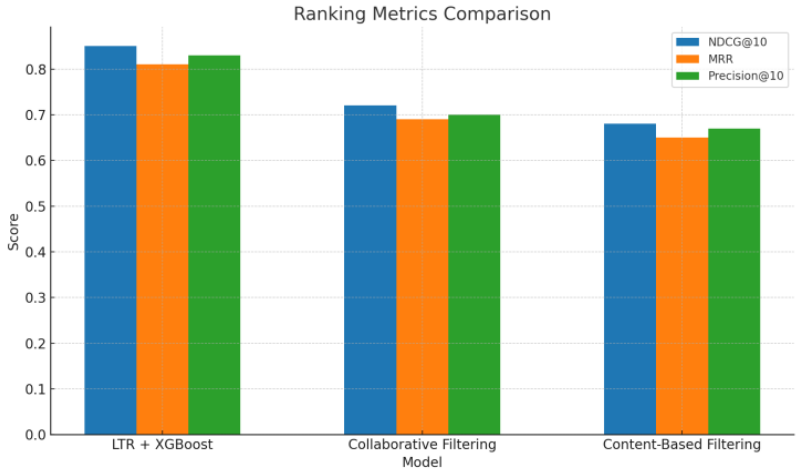


Figure 2: comparison of performance across all metrics.

Effect of the Feedback Loop

The role of the feedback mechanism was examined by measuring model performance in two operational modes: one that leveraged real-time user interactions (clicks, enrollments, ratings), and the other that operated statically without feedback. Incorporating dynamic feedback significantly enhanced recommendation quality, with NDCG@10 improving by 9% (from 0.78 to 0.85), and MRR increasing by 11% (from 0.73 to 0.81). These improvements demonstrate the model's ability to dynamically adjust rankings in response to evolving user preferences, supporting prior findings by Zhang et al. (2019) and Joachims (2002) that user-interaction data play a critical role in refining model personalisation and responsiveness. This adaptive behaviour aligns with the principles discussed by Brusilovsky and Millán (2007), who emphasised that real-time adaptability is vital in educational recommender systems to sustain learner engagement.

Table 2: Performance Impact of Feedback Loop Integration

Metric	Without Feedback	With Feedback
NDCG@10	0.78	0.85
MRR	0.73	0.81
Precision@10	0.75	0.83

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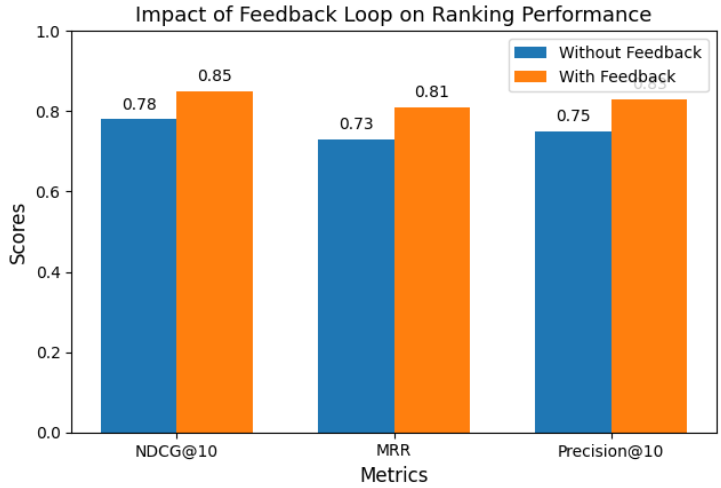


Figure 3: Comparison of performance with feedback

Beyond the internal comparison of the model with and without feedback, the final feedback-enhanced LTR model was also evaluated against traditional baselines such as Collaborative Filtering (CF) and Content-Based Filtering (CBF). The LTR model with feedback consistently outperformed both baselines across all primary metrics. For instance, NDCG@10 improved by 18% over CF and 25% over CBF, while MRR increased by 17% and 24%, respectively. These results corroborate earlier research by Su and Khoshgoftaar (2009) and Resnick and Varian (1997), who noted the limitations of CF and CBF in handling evolving user behaviour and contextual relevance. The demonstrated performance gains further reinforce the findings of Qin et al. (2010), who showed that Learning-to-Rank models can directly optimise ranking metrics and outperform heuristic-based recommendation strategies.

Overall, the results shown in Table 2 and Figure 3 confirm that integrating Learning-to-Rank with XGBoost provides a robust, scalable, and high-precision framework for personalised course recommendations. The substantial improvements over collaborative and content-based baselines indicate that the model effectively captures complex relationships between user queries, course metadata, and behavioural feedback. Furthermore, the incorporation of a real-time feedback loop ensures continuous model refinement, echoing the conclusions of Tang and McCalla (2005) that adaptive, feedback-driven recommendation systems are essential for maintaining learner engagement and improving user satisfaction in dynamic e-learning environments.

CONCLUSION

This study presented a robust and adaptive Learning-to-Rank (LTR) framework integrated with the XGBoost algorithm to deliver real-time, personalised course recommendations in dynamic e-learning environments. By embedding a real-time feedback loop, the proposed system continuously refines its ranking model based on user interactions such as clicks, enrollments, and ratings, enabling it to adapt seamlessly to evolving learner preferences. Leveraging a diverse

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dataset collected from major MOOC platforms, including Coursera, Udemy, and edX, the framework effectively models complex user–course relationships using TF-IDF representations, user embeddings, and engagement-driven features for fine-grained ranking.

Experimental results demonstrated that the proposed feedback-enhanced LTR model substantially outperforms traditional collaborative filtering and content-based approaches across all key performance metrics, achieving an NDCG@10 of 0.85, MRR of 0.81, and Precision@10 of 0.83, while maintaining a low average latency of 112 milliseconds. These findings confirm that integrating LTR with XGBoost not only enhances ranking precision but also ensures real-time responsiveness—an essential attribute for large-scale educational platforms.

The feedback mechanism proved particularly influential, yielding measurable improvements in adaptability and user satisfaction, thereby reinforcing conclusions from prior research that highlight the importance of interaction-driven learning in recommendation systems. Overall, the research highlights the importance of integrating advanced ranking algorithms with continuous user feedback to deliver accurate, context-aware, and scalable course recommendations. The proposed framework provides a generalizable foundation for enhancing user engagement and learning outcomes across varied educational contexts.

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